Chapter 14

Basic Op-Amp Circuits & Applications

14.1 Introduction

The Operational Amplifier (Op-Amp) is arguably the most versatile component in the analog electronics arsenal. While Chapter 13 focused on the internal characteristics of the Op-Amp (Gain, Impedance, Slew Rate), this chapter focuses on **Applications**.

By applying **Negative Feedback**—feeding a portion of the output back to the inverting input—we can trade the Op-Amp's excessively high, unstable open-loop gain for a precise, stable, closed-loop gain determined solely by external resistors.

This chapter covers the linear applications essential for agricultural instrumentation, signal processing, and control systems.

Learning Objectives

By the end of this chapter, you should be able to:

- Analyze Inverting and Non-Inverting configurations using the Virtual Ground concept.
- Explain the operation of a Voltage Follower (Buffer) and its importance in impedance matching.
- Derive the output equation for Summing Amplifiers (Adders).
- Analyze Difference Amplifiers and Instrumentation Amplifiers.
- Describe the function of Integrator and Comparator circuits.

14.2 The Golden Rules of Analysis

To simplify the analysis of any Op-Amp circuit with negative feedback, we apply two fundamental assumptions known as the "Golden Rules". These are based on the properties of an Ideal Op-Amp.

1. **The Voltage Rule:** The output attempts to do whatever is necessary to make the voltage difference between the two inputs zero.

$$V_+ = V_- \tag{14.1}$$

2. **The Current Rule:** The inputs draw no current.

$$I_+ = I_- = 0 \tag{14.2}$$

14.3 The Inverting Amplifier

The Inverting Amplifier is the most common Op-Amp configuration. It amplifies the signal and inverts its phase (180° shift).

14.3.1 Circuit Construction

- The Non-Inverting input (+) is grounded.
- The input signal V_{in} is applied to the Inverting input (-) through a resistor R_{in}.
- A feedback resistor R_f connects the output to the Inverting input.

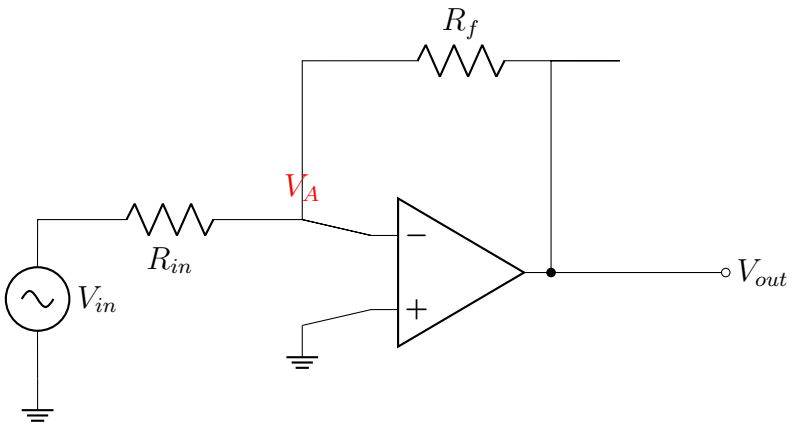


Figure 14.1: The Inverting Amplifier Circuit.

14.3.2 Mathematical Derivations

Let voltage at the inverting node be V_A.

1. Since $V_+ = 0$V (Ground), by the Voltage Rule, $V_- = V_A = 0$V. This node is a **Virtual Ground**.
2. Apply KCL at node V_A:

   $$I_{in} = I_f$$

   (Current flowing in from source must flow out through feedback resistor, as no current enters the Op-Amp).
3. Write currents in terms of voltages:

   $$\frac{V_{in} - V_A}{R_{in}} = \frac{V_A - V_{out}}{R_f}$$

4. Since $V_A = 0$:

   $$\frac{V_{in}}{R_{in}} = \frac{-V_{out}}{R_f}$$

5. Solving for Voltage Gain ($A_v = V_{out}/V_{in}$):

$$A_v = -\frac{R_f}{R_{in}} \tag{14.3}$$

14.3.3 Characteristics

- **Gain:** Determined purely by resistor ratio. Can be > 1 or < 1.
- **Input Impedance (Z_{in}):** Because of the Virtual Ground, the input source "sees" R_{in} connected to ground.

  $$Z_{in} = R_{in}$$

  Disadvantage: If high gain is needed (R_f high), R_{in} must be low, which loads the source.

Example 1: Inverting Amplifier Design

Design an amplifier with a gain of -10 and an input impedance of $10k\Omega$.

Solution: 1. Set $Z_{in} = R_{in} = 10k\Omega$. 2. Use gain formula:

$$A_v = -\frac{R_f}{R_{in}} \implies -10 = -\frac{R_f}{10k\Omega}$$

3. Solve for R_f:

$$R_f = 10 \times 10k\Omega = 100k\Omega$$

14.4 The Non-Inverting Amplifier

This configuration provides high input impedance and zero phase shift.

14.4.1 Circuit Construction

- The input signal V_{in} is applied directly to the Non-Inverting (+) input.
- Feedback is applied to the Inverting (-) input via a voltage divider (R_f and R_{in}).

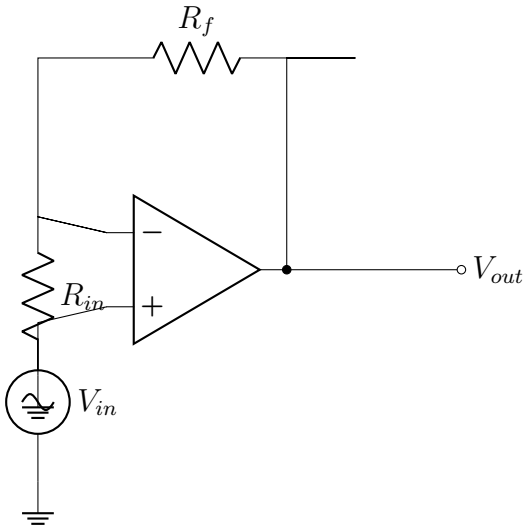


Figure 14.2: The Non-Inverting Amplifier Circuit.

14.4.2 Mathematical Derivations

1. Since $V_+ = V_{in}$, by the Voltage Rule, $V_- = V_{in}$.

2. The voltage at the inverting node V_- is determined by the voltage divider formed by R_f and R_{in} acting on V_{out}:

$$V_- = V_{out}\left(\frac{R_{in}}{R_{in} + R_f}\right)$$

3. Equate V_- to V_{in}:

$$V_{in} = V_{out}\left(\frac{R_{in}}{R_{in} + R_f}\right)$$

4. Rearrange for Gain:

$$A_v = \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_{in}} \tag{14.4}$$

14.4.3 Characteristics

- **Gain:** Always ≥ 1. Cannot attenuate signals.
- **Input Impedance (Z_{in}):** Extremely High ($\approx \infty$). This is ideal for measuring sensors without loading them.

14.5 The Voltage Follower (Buffer)

A special case of the Non-Inverting amplifier where $R_f = 0$ (Short) and $R_{in} = \infty$ (Open).

14.5.1 Characteristics

Using the Non-Inverting formula:

$$A_v = 1 + \frac{0}{\infty} = 1$$

$$V_{out} = V_{in}$$

Why use a circuit with a gain of 1?

- **Impedance Matching:** It transforms a High Impedance source to a Low Impedance output.
- It acts as a **Buffer** between a delicate sensor and a heavy load.

14.6 Summing Amplifier (Adder)

The Summing Amplifier allows us to add multiple analog signals together mathematically. It is based on the Inverting configuration.

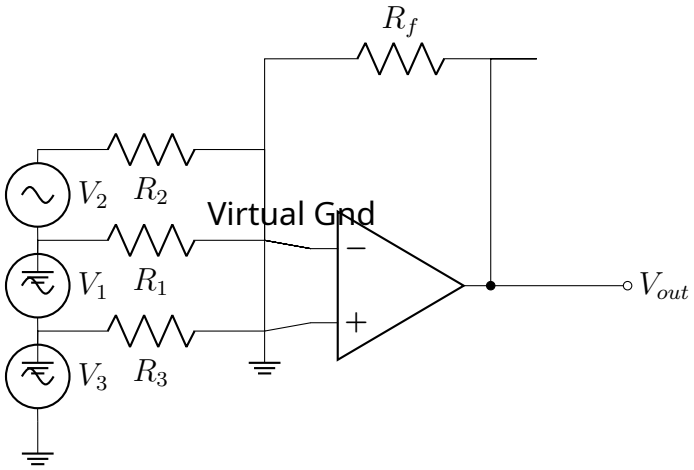


Figure 14.3: Summing Amplifier with 3 Inputs.

14.6.1 Derivation

At the Virtual Ground node, KCL applies:

$$I_1 + I_2 + I_3 = I_f$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_{out}}{R_f}$$

Rearranging:

$$V_{out} = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right) \tag{14.5}$$

If all resistors are equal ($R_1 = R_2 = R_3 = R_f$):

$$V_{out} = -(V_1 + V_2 + V_3)$$

14.7 Difference Amplifier (Subtractor)

This circuit amplifies the difference between two inputs while rejecting any common signal (utilizing the Op-Amp's CMRR).

14.7.1 Analysis

By applying Superposition theorem:

$$V_{out} = \left(\frac{R_f}{R_{in}}\right)(V_2 - V_1) \tag{14.6}$$

(Assuming perfectly matched resistor ratios).

14.8 The Integrator

By replacing the feedback resistor with a Capacitor (C), the Op-Amp performs the mathematical operation of Integration (calculus).

14.8.1 Circuit and Derivation

Current through input resistor R:

$$I_{in} = \frac{V_{in}}{R}$$

Current through feedback capacitor C (Recall $I = C\frac{dV}{dt}$):

$$I_f = C\frac{d(0 - V_{out})}{dt} = -C\frac{dV_{out}}{dt}$$

Equating currents ($I_{in} = I_f$):

$$\frac{V_{in}}{R} = -C\frac{dV_{out}}{dt}$$

Integrating both sides:

$$V_{out} = -\frac{1}{RC}\int V_{in}\,dt \tag{14.7}$$

14.8.2 Waveform Response

If the input is a constant DC voltage (Step function), the integral is a Ramp. If the input is a **Square Wave**, the integral is a **Triangle Wave**.

14.9 Comparators

A Comparator operates in **Open Loop** mode (no feedback). It compares an input voltage to a reference.

- If $V_{in} > V_{ref}$, $V_{out} \to +V_{sat}$ ($+V_{CC}$).
- If $V_{in} < V_{ref}$, $V_{out} \to -V_{sat}$ ($-V_{EE}$).

This produces a digital (High/Low) output from an analog input.

14.10 Chapter Summary

1. **Inverting Amp:** $Gain = -R_f/R_{in}$. Input impedance is R_{in}.
2. **Non-Inverting Amp:** $Gain = 1 + R_f/R_{in}$. Input impedance is infinite.

3. **Buffer:** Gain = 1. Used for isolation.
4. **Summing Amp:** Adds voltages with weighing factors.
5. **Integrator:** Converts Square wave to Triangle wave.
6. **Comparator:** Digital output based on analog comparison.

14.11 Review Questions

Part A: Conceptual Questions

1. Explain the concept of ”Virtual Ground”. Why does it exist in the Inverting amplifier but not the Non-Inverting amplifier?
2. Why is a Voltage Follower (Gain=1) useful if it doesn't amplify the signal voltage?
3. Describe the difference between negative feedback and positive feedback. Which one is used for amplifiers and which for oscillators/comparators?
4. What happens to the output of an Integrator if the input is a constant positive DC voltage?
5. Why are Integrator circuits often placed with a resistor in parallel with the capacitor? (Hint: DC Gain).
6. Can an Op-Amp output a voltage higher than its power supply rails? Explain.
7. Draw the circuit of a Summing Amplifier that produces $V_{out} = -(V_1 + V_2 + V_3)$.
8. Ideally, what is the input impedance of a Non-Inverting Amplifier?
9. Why do we use a resistor R_{comp} equal to $R_f||R_{in}$ on the non-inverting terminal of an inverting amplifier?
10. Explain the function of a Comparator in an analog-to-digital converter.

Part B: Advanced Calculations & Analysis

11. **Inverting Design:** Design an inverting amplifier with a gain of -50 and an input impedance of at least $2k\Omega$. Specify R_{in} and R_f.
12. **Non-Inverting Analysis:** A non-inverting amplifier has $R_{in} = 4.7k\Omega$ and $R_f = 47k\Omega$.
    (a) Calculate the Voltage Gain (A_v).
    (b) If $V_{in} = 0.5$ V, what is V_{out}?

(c) If the Op-Amp saturates at ± 13V, what is the maximum input voltage allowed?

13. **Summing Amplifier:** A summing amp has $R_f = 100k\Omega$. It has three inputs with resistors $R_1 = 10k\Omega$, $R_2 = 20k\Omega$, and $R_3 = 50k\Omega$. Inputs: $V_1 = 0.2$V, $V_2 = -0.5$V, $V_3 = 0.8$V. Calculate the output voltage.

14. **Integrator:** An integrator has $R = 100k\Omega$ and $C = 1\mu F$. The input is a square wave with peak amplitude 2V and frequency 1 kHz. Calculate the peak-to-peak amplitude of the output triangle wave. *(Hint: Period $T = 1ms$. Integration time is $T/2$. $V_{out} = \frac{1}{RC} \times V_{in} \times \frac{T}{2}$).*

15. **Current-to-Voltage Converter:** A photodiode produces a current of $5\mu A$. Design an Op-Amp circuit to convert this current into an output voltage of -2V. Draw the circuit and calculate the feedback resistor value.